

Project Design Phase-II Technology Stack (Architecture & Stack)

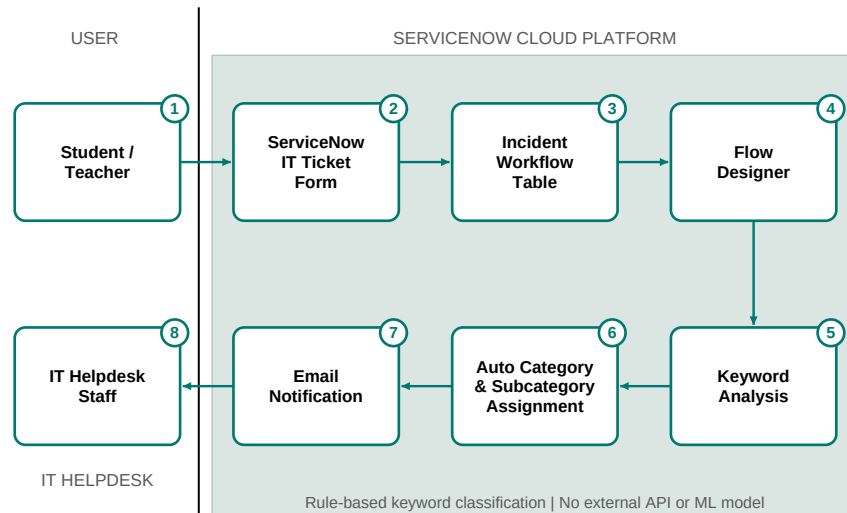
Date	30 September 2026
Team ID	SWTID-2026-2046
Project Name	Auto Ticket Classification using Flow Designer
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Auto Ticket Classification using Flow Designer (ServiceNow School IT Helpdesk)

Reference: <https://docs.servicenow.com>



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

S.No	Component	Description	Technology
1.	User Interface	Interface used by students and teachers to create IT support tickets and enter issue details.	ServiceNow Ticket/Form Interface

2.	Application Logic-1	Automatically analyzes the ticket short description and identifies predefined issue keywords.	ServiceNow Flow Designer
3.	Application Logic-2	Automatically assigns the appropriate Category and Subcategory based on the identified issue.	Flow Designer If / Else If Conditions and Update Record Actions
4.	Application Logic-3	Sends an automated email notification to the caller after ticket creation and classification.	ServiceNow Flow Designer Send Email Action
5.	Database	Stores structured IT ticket information including caller, category, subcategory, description, state, assigned group and assigned user.	ServiceNow Incident Workflow / Custom Table
6.	Cloud Database	Not separately used in the project because the solution uses the ServiceNow platform for data storage.	ServiceNow Platform Database
7.	File Storage	Stores project/update-set export files when required for deployment or transfer.	ServiceNow Update Set XML / Local File Storage
8.	External API-1	No external API is required for the ticket classification process.	Not Used
9.	External API-2	No external API is required for the ticket classification process.	Not Used
10.	Machine Learning Model	The project uses predefined keyword-based Flow Designer conditions instead of a machine learning model.	Not Used — Rule-Based Classification
11.	Infrastructure (Server / Cloud)	The application is developed and executed within the ServiceNow platform. Local Server Configuration: Not required Cloud Server Configuration: ServiceNow cloud instance	ServiceNow Cloud Platform

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	The solution is implemented using ServiceNow Flow Designer and does not depend on external open-source frameworks.	ServiceNow Flow Designer
2.	Security Implementations	Access to ticket information and modification of records should be restricted to authorized users and appropriate ServiceNow roles.	ServiceNow Roles and Access Controls
3.	Scalable Architecture	The solution can be extended by adding new categories, subcategories, keywords, automation rules and additional workflow actions without redesigning the entire system.	ServiceNow Flow Designer
4.	Availability	The solution operates on the ServiceNow platform, allowing users to create and manage IT support tickets through the available ServiceNow environment.	ServiceNow Cloud Platform
5.	Performance	Automatic classification occurs when a ticket is created, reducing manual classification work and improving the efficiency of ticket routing.	ServiceNow Flow Designer Automation

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>